

CH230: Problem 2

November 13, 2025

A chemical species A from a gas layer diffuses into a liquid layer where it is subjected to an irreversible first-order reaction that converts it to a product B . Steady-state mass balance can be used to derive the following ordinary differential equations for compound A :

$$\frac{d}{dx} \left(D \frac{dC_A}{dx} \right) - kC_A = 0, \quad 0 < x < L$$

where D is the diffusion coefficient and varies linearly ($D = 0.84 \text{ cm}^2/\text{day}$ at $x = 0$ and $D = 1.09 \text{ cm}^2/\text{day}$ at $x = L$), and k is the first-order rate constant for the conversion of A to B ($k = 0.1/\text{day}$).

The following boundary conditions hold:

$$C_A = 100 \text{ mol/L} \quad \text{at } x = 0,$$

$$\frac{dC_A}{dx} = 0 \quad \text{at } x = L.$$

Use the finite-difference method [$\mathcal{O}((\Delta x)^2)$] to compute the steady-state concentration of A from $x = 0$ to L , where $L = 5 \text{ cm}$. Take $\Delta x = 0.05 \text{ cm}$. Fit a natural cubic spline through the numerical solution obtained at the grid points. Through a plot, compare (i) the numerical solution with the analytical solution and (ii) the interpolated solution with the analytical solution.